DEEP-TECH MARKET RESEARCH

Physical AI in Switzerland and Europe

An opinionated market map

Lucas Bergerot

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3 documents

Contents

PART 1

The Market Map

Generated 2026-08-18 from 5 scored companies.

This ranks companies building robots and autonomous machines that perceive and act in the physical world. The criteria and their weights are the argument; the scores are its consequence. Both are stated in `00-scoping.md` and were fixed before any company was scored.

The weighting takes two positions worth disagreeing with: traction is redefined as units deployed at paying customers rather than money raised, and timing is capped at 10 because the "why now" of Physical AI is identical for every company here.

1. The ranking

#	Company	Score	Hand-score	Gap	Lowest-confidence criterion
1	**ANYbotics**	**86/100**	93/100	-7	Market (MEDIUM)
2	**Verity**	**85/100**	92/100	-7	none below HIGH
3	**Gravis Robotics**	**70/100**	85/100	-15	none below HIGH
4	**Humanoid**	**59/100**	77/100	-18	none below HIGH
5	**mimic robotics**	**54/100**	60/100	-6	Market (MEDIUM)

2. Calibration — the model against a blind hand-score

Five companies were scored by hand before any pipeline code existed. The model never saw those scores. The comparison is the point of this project: a ranking nobody has checked is an opinion with extra steps.

Disagreement per criterion, in notches out of 5. A negative number means the model scored lower than the hand-score.

Company	Field traction	Team / execution	Technology	Market	Timing	Total gap
ANYbotics	—	—	-1	-1	—	**-7**
Verity	—	—	-1	-1	—	**-7**
Gravis Robotics	-1	—	-2	-1	+1	**-15**
Humanoid	-2	—	—	-2	—	**-18**
mimic robotics	—	-1	-1	+1	—	**-6**
Sum	**-3**	**-1**	**-5**	**-4**	**+1**	

Agreement: 12 of 25 criterion scores identical, 22 of 25 within one notch.

Rank order: identical on both sides.

Hand: ANYbotics > Verity > Gravis Robotics > Humanoid > mimic robotics

Model: ANYbotics > Verity > Gravis Robotics > Humanoid > mimic robotics

The argued verdict on each disagreement is in `00-scoping.md` §5.

3. The companies

1. ANYbotics — 86/100

Switzerland · legged robots for industrial inspection

Criterion	Score	Weight	Points	Confidence	Hand-score
Field traction	5/5	30	30.0	HIGH	5/5 (M)
Team / execution	4/5	25	20.0	HIGH	4/5 (H)
Technology	4/5	20	16.0	HIGH	5/5 (H)
Market	4/5	15	12.0	MEDIUM	5/5 (H)
Timing	4/5	10	8.0	HIGH	4/5 (H)

Why these scores

Field traction. This clears the top anchor on customer-side evidence, not just company claims. Outokumpu's own newsroom confirms three named ANYmal robots in daily service across three plants in Germany, Sweden and Finland, with a 2023 deal expanded in 2024 — that is one customer past pilot into a multi-site fleet with a repeat order. PETRONAS' own press release confirms a 2019 co-development moving to a signed commercial agreement in March 2022 to scale deployment. GE Vernova's own site describes a two-week PoC converting into deployment at a customer site in Ireland, and Equinor put an ANYmal D into the Northern Lights CCS facility integrated into its own Flotilla fleet software. Vendor/investor claims of ~200 units shipped and thousands of weekly inspections are company-sourced and discounted, but even without them the customer-side accounts alone establish repeat orders and units in daily service across multiple named industrials. A 3 would require each install to be custom with only 3-10 customers; the named roster (BP, Equinor, Novelis, Outokumpu, Equans, BASF, DSM-Firmenich, Grace) plus SLB as channel partner and RaaS commercial packaging exceeds that. Sources: 1, 2, 3, 4, 5, 6

Team / execution. The scale explicitly caps ETH/EPFL pedigree at 3, and this is the archetypal ETH RSL spin-off (Fankhauser, Hutter, Fässler). What pushes above 3 is the multi-generation shipping record rather than the pedigree: ALoF (2009) to StarlETH (2012) to ANYmal Alpha (2015) to commercial ANYmal C deliveries from 2020, ANYmal D as fourth generation, and the separately certified ANYmal X unveiled 2022 with commercial orders in 2023 — that is well past 'one generation shipped.' Hiring has scaled from ~100 to ~200-260 across sources, and it has pulled a Chief Product Officer with May Mobility, Intel and Qualcomm background, evidencing senior big-tech draw. It does not reach 5 because no founder has shipped hardware at scale before, generation cycles are long (2015 prototype to 2020 first deliveries, ANYmal X commercial launch still targeted for 2026 four years after unveiling) rather than shortening, and headcount figures are irreconcilable. Funding (~$150M cumulative) counts here only as resourcing, and it is substantial. Sources: 1, 2, 3, 4, 5, 6

Technology. The decisive item for above-3 is the ATEX/IECEx Zone 1 IIB certification on ANYmal X — a multi-year certification path (co-development from 2019, unveiled 2022, commercial launch targeted 2026) that is exactly the 'multi-year certification' moat named in the 5 anchor, and it is a hard barrier competitors cannot shortcut. Supporting depth: the proprietary ANYdrive actuator (frameless motor, harmonic reducer, planar spiral spring) now industrialized with maxon, IP67 rating, and an AWS-based fleet backend deploying identical containers across sites, with a Data Navigator analytics layer that processed 2,500+ inspections in two weeks across Grace, Outokumpu and DSM-Firmenich — the beginnings of a data flywheel. It stops short of 5 because the IP position is thin and partly borrowed: only 8 patents filed per CB Insights, and a reported patent licensing fee paid to Boston Dynamics in 2019; the actuator is now produced by an external supplier, and the inspection-data asset is only described in pilot terms rather than demonstrated to compound into model performance. Above 3 because the certification is not 'broadly available.' Sources: 1, 2, 3, 4, 5, 6, 7

Market. The buyers are named and they are already purchasing in this category: BP, Equinor, Petrobras, ENI, PETRONAS, Shell, SLB in oil & gas; BASF, Borealis, DSM-Firmenich, Grace in chemicals; Outokumpu and Novelis in metals; GE Vernova, Siemens Energy and Swiss nuclear KKL/Axpo in power. Inspection and asset-integrity is an existing, budgeted line item at these operators — Outokumpu framed it under safety management, and GE Vernova folded it into its own APM software, evidence of an established procurement slot rather than a speculative new one. SLB as 'preferred ground robotics supplier' gives a channel into that budget, and RaaS lowers capex friction. It falls short of 5 because the segment is visibly crowded — ANYbotics itself markets against Boston Dynamics' Spot, and Unitree and others are listed as competitors — and heavy-industry/oil-major procurement cycles are long (PETRONAS 2019 engagement to 2026 full commercial launch), which is precisely the starving-cycle risk in the 3 anchor. Above 3 because the pain (Ex-zone human inspection) is urgent and safety-driven with buyers already spending. Sources: 1, 2, 3, 4, 5, 6

Timing. There is a dateable unlock, but it is one ANYbotics manufactured rather than one handed to it by the outside world. The ATEX/IECEx Zone 1 certification of ANYmal X — co-development from 2019, public unveiling 22 March 2022 at OTC Asia, Early Adopter Program 2022, commercial ordering 2023 — is a specific, dated regulatory threshold that opens Zone 1 hazardous areas that were previously closed to any legged robot, and it is what makes the oil & gas and chemicals pipeline addressable now. The Climate Investment round of September 2025, from a fund owned by OGCI members (Aramco, BP, Chevron, Equinor, ExxonMobil, Shell), also points to a customer-side decarbonization/emissions-monitoring mandate driving demand at a datable moment. It is not a 5 because the certification is the company's own achievement on a schedule it set, with full commercial launch still pushed to 2026, and no external regulation or component price threshold is shown compelling buyers to act by a given date; the 2022 anti-weaponization pledge is explicitly not a regulation. It is above the default 3 because the evidence names a concrete, dated gating event rather than merely riding the general Physical AI wave. Sources: 1, 2, 3, 4

Contradictions in the sources, left unresolved

- Headcount figures are inconsistent across sources and dates, ranging from 'more than 100' (older LinkedIn copy) to 'more than 190' (2025 job posting) to 'over 200 experts' (Dec 2024 release) to 251 (Tracxn, March 2026) to 260 (RocketReach) to Wikipedia's static '150'; these are not reconcilable without knowing each source's snapshot date.
- Unit-count and reservation figures are muddled: '500 ANYmal X quadrupeds reserved' (2023, a pre-order figure) versus 'over 200 units... shipped' and 'approximately 200 ANYmal robots... deployed' (2024-2025 releases); unclear whether the ~200 figure includes only ANYmal/ANYmal D or also ANYmal X, and whether 'shipped' equals 'actively deployed.'

- Funding totals vary by currency and rounding: reports cite '$130 million,' 'over $150 million,' and 'over €127 million' as the cumulative total at different points in 2024-2025, without a single reconciled ledger.
- All specific unit/fleet figures trace back to ANYbotics or its investors' own statements, even when repeated by trade press; the strongest independent confirmations found are the Outokumpu, PETRONAS, Equinor, and GE Vernova customer-side accounts, but none of these publish a company-wide fleet total.
- Patent depth is thin: only '8 patents' filed per CB Insights, and one historical account notes ANYbotics paid Boston Dynamics a patent licensing fee in 2019, suggesting partial reliance on external IP; no primary patent filings were reviewed directly to confirm scope or currency of this.

2. Verity — 85/100

Switzerland · autonomous indoor drones for warehouse inventory tracking

Criterion	Score	Weight	Points	Confidence	Hand-score
Field traction	5/5	30	30.0	HIGH	5/5 (H)
Team / execution	5/5	25	25.0	HIGH	5/5 (H)
Technology	3/5	20	12.0	HIGH	4/5 (M)
Market	4/5	15	12.0	HIGH	5/5 (H)
Timing	3/5	10	6.0	HIGH	3/5 (M)

Why these scores

Field traction. This goes well past the 3-anchor of "3-10 customers, each install custom." IKEA/Ingka deploying 100 drones across 16 European locations, KeHE starting at one Arizona site in summer 2023 and then rolling out across US distribution centers (the Portland site being cited as the 100th installation) is a textbook repeat order and pilot-to-fleet expansion; DSV runs Verity across multiple sites including a 10,000+ pallet-location facility, and On has a fleet scanning daily at a US plant. Named, customer-attributed quotes exist from DSV, KeHE and On, not just Verity's own count. It falls short of nothing in the anchor; the counts (30 sites 2023 → 100+ 2024 → 150+ 2025) are company-sourced and one Maersk pilot site is confirmed dead, which is why this is a 5 on the strength of the customer-confirmed fleet expansions rather than on the headline numbers alone. Sources: 1, 2, 3, 4, 5, 6, 7

Team / execution. The ETH/EPFL spinout pedigree alone would cap at 3, but co-founder Raffaello D'Andrea also co-founded Kiva Systems, which shipped warehouse robots at scale and was acquired by Amazon in 2012 — prior hardware shipped at scale, which is exactly the 5 anchor. Beyond that there is generational evidence: a live-events drone business (30,000+ autonomous flights over people, 7,000 incident-free at Cirque du Soleil) preceded the warehouse product launched in 2020, and the FCC filing shows the company now on a Series 4 indoor system. Hiring is not flat or purely academic — field, manufacturing, test, refurbishment and contract-manufacturing quality roles are open, and senior hires include an ex-Covariant business development head; ~80-100+ staff. This exceeds 4 because both conditions of the top anchor (prior at-scale hardware and multiple shipped generations) are evidenced, not just one. Sources: 1, 2, 3, 4, 5, 6

Technology. Above 1 clearly: the drones are purpose-built rather than commodity hardware (The Robot Report explicitly contrasts Verity's custom airframes with Gather AI's off-the-shelf approach), GPS-denied SLAM navigation runs unattended ceiling-to-floor scans off-hours at claimed 99.9% accuracy, and there is genuine control-theory depth (learned quadcopter control from Hehn's ETH doctorate; a propulsion-failure safe-landing algorithm). But it does not reach 4-5 because the approach is demonstrably available to others — the same market contains Gather AI, Corvus, EyeSee Drones, B Garage and inventAIRy XL doing indoor inventory drones, and the safe-landing algorithm is described as something that "can be integrated in all quadrotors," i.e. portable rather than unbuyable. No evidence of a proprietary data flywheel that compounds with each scan, and the only certification cited (FCC Covered List conditional approval) is time-limited to end-2026 rather than a multi-year moat. The RFID noise-rejection add-on is a differentiator but is described only by Verity's own CRO. Sources: 1, 2, 3, 4, 5

Market. Better than the 3 anchor because the pain is already budgeted and bought: DSV's operations manager frames the drones against contractual SLA requirements for specific order types, i.e. an operational necessity, and the buyer set spans 3PL (DSV, UPS SCS, Maersk/Performance Team), retail (Ingka/IKEA), food distribution (KeHE), CPG (Haleon) and footwear manufacturing (On) — these are firms that purchase inventory-accuracy tooling today, and KeHE's multi-DC rollout shows procurement cycles that are not startup-starving. Held back from 5 by crowding that the evidence itself documents: at least five direct competitors (Gather AI, Corvus, EyeSee, B Garage, inventAIRy XL) and an ongoing consolidation event in which Gather AI absorbed Ware Robotics — a sign the category is contested and possibly not yet large enough to support all entrants. Sources: 1, 2, 3, 4, 5

Timing. No specific dateable unlock explains why now. The one regulatory fact that is genuinely permissive — FAA Part 107 not applying to indoor-only operations — has been true throughout and is a standing condition, not a change; indeed the company was founded in 2014 and launched the warehouse product in 2020, so the enabling environment long predates the present. The dated FCC/Pentagon Covered List item is a conditional import/marketing approval expiring 31 December 2026, which is a compliance hurdle Verity had to clear as a non-US manufacturer rather than a market-opening mandate. The cited "since 2019 indoor inventory drones reshaped industrial drone use" is exactly the general Physical AI/warehouse-automation wave that the default anchor covers. Above 1 because the company did not sit on an obvious opportunity — the 2019-2020 emergence of the indoor inventory-survey category is real and Verity launched into it — but nothing here earns 4 or 5. Sources: 1, 2, 3

Contradictions in the sources, left unresolved

- LinkedIn's own company page states 'more than 100' employees, while a third-party startup profile states 'approximately 80 people.' Neither figure is independently audited.
- Deployment figures cited over time (30 sites in March 2023, 80 warehouses in 2023/2024, 100 customer warehouses/sites in Aug-Sept 2024, 150 global deployments in Sept 2025) all trace back to Verity's own statements, even when repeated by trade press; no independent third party has published its own unit count.
- Tracxn lists the March 2023 Series B tranche as '$38.8M' while Verity's own press release and multiple wire outlets state '$32M' (30M CHF) for that same tranche, with an additional $11M in July 2023 — a discrepancy of roughly $7M.
- Verity/trade press cite Maersk as a flagship logo, but Supply Chain Dive's independent reporting is the only source noting that the original Miami pilot site is confirmed by Maersk to be 'no longer

operational'; later Maersk references (e.g., in RFID pilot news, 2025 podcast) do not clarify whether these are new/different sites.

3. Gravis Robotics — 70/100

Switzerland · autonomy retrofit kits for heavy construction machinery

Criterion	Score	Weight	Points	Confidence	Hand-score
Field traction	3/5	30	18.0	HIGH	4/5 (H)
Team / execution	4/5	25	20.0	HIGH	4/5 (H)
Technology	3/5	20	12.0	HIGH	5/5 (H)
Market	4/5	15	12.0	HIGH	5/5 (H)
Timing	4/5	10	8.0	HIGH	3/5 (H)

Why these scores

Field traction. The evidence reaches named, multi-country counterparties beyond pure demos: Holcim invested and expanded Gravis from construction into quarry work explicitly 'following a successful UK pilot', Kibag in Switzerland is running Gravis kit on Develon machines, and Flannery Plant Hire has a distribution-style arrangement plus the $8M CAM Pathfinder project. That is more than the 1-anchor (demos/pilots/LOIs). But it does not reach 5: there is no evidence of repeat orders, no unit counts, no customer described as past pilot into a fleet. The most concrete recent items are still framed as trials and shows — a 'UK-first trial' of an autonomous excavator at Manchester Airport (Oct 2025), Bauma 2025 live demonstrations, and a planned CONEXPO 2026 Hitachi demo. The 'seven countries / four continents' and 'dozens of brands' claims are company statements relayed by press, not third-party confirmation, and no paying-customer count or revenue is confirmed by a primary source (only an aggregator's unverified $10.1M). Roughly 3-10 named accounts, each install custom per machine brand, is exactly the 3 anchor. Sources: 1, 2, 3, 4, 5, 6, 7

Team / execution. This is an ETH Zurich spin-off, which the rubric caps at 3 on pedigree alone — Johns and Jud come out of the HEAP walking-excavator programme, with Marco Hutter (DARPA SubT winner) on the board. What lifts it above 3 is shipped product iteration plus extraordinary resources: the Rack retrofit kit went from first work with Menzi Muck to OEM-revealed integrations with CASE, Develon, Hitachi, Menzi Muck, Sumitomo and Yanmar at Bauma April 2025, followed by a second-generation 'Copilot' system debuted at CONEXPO March 2026 — two generations inside about eighteen months. Capital went from a $23m round (Nov) to a $200m SoftBank Series A at $1bn post-money (Aug 2026), a statement about resources to hire and scale. It does not reach 5: no founder has shipped hardware at volume before, the founders are academics and an architect by background, headcount is unconfirmed (26 vs 55 across two aggregators), and there is no evidence of pulling senior talent out of big tech. Sources: 1, 2, 3, 4, 5, 6

Technology. The Rack is a genuinely engineered retrofit stack — five surround cameras, lidar, onboard compute, GNSS, plus the Slate 3D tablet interface — backed by nearly a decade of ETH heavy-machinery autonomy research, and the learning-based controller that infers soil state from hydraulic pressure, vibration and engine strain is more than a bolt-on perception kit. That places it well above 1. It does not reach 5 because nothing in the evidence shows compounding: no proprietary data flywheel is documented (only a company description of the control approach), no unbuyable component (cameras, lidar, GNSS are commodity), and no multi-year certification barrier. BuiltWorlds explicitly places Gravis alongside Hive Autonomy and SafeAI doing retrofittable autonomy on heavy machinery, i.e. the approach is broadly available. The CEO's own framing — that OEMs' move to electronic joystick signals makes it easy to plug in a retrofit computer — cuts against defensibility, since the same door opens for competitors. Sources: 1, 2, 3, 4, 5

Market. The buyers named are organisations that already purchase in this category: Holcim (quarrying), Taylor Woodrow and Morgan Sindall (contractors), and Flannery Plant Hire, the UK's largest equipment rental provider, whose entire business is putting machines on sites — retrofit autonomy sits on an existing plant-hire budget line rather than requiring a new one. The autonomous construction equipment market is independently sized at $8.8bn in 2023 growing >7.5% annually (Global Market Insights via Fortune), and the UK's $716bn infrastructure pipeline is a real demand backdrop. It falls short of 5 because the segment is contested by comparable retrofit players (SafeAI, Hive Autonomy per BuiltWorlds) and because construction/quarry procurement runs through slow contractor and rental-fleet cycles; the $1.6 trillion 'earthmoving industry' figure is a company TAM claim, not a served market. Above 3 because the pain is budgeted and the buyers are already named and transacting, not merely addressable. Sources: 1, 2, 3, 4, 5, 6

Timing. Above the default 3 because two concrete, company-specific unlocks are documented rather than the generic Physical AI wave. First, the CEO identifies a technical threshold: OEMs' industry-wide switch from hydraulic pilot-stage joysticks to electronic joystick signals, which is what makes a plug-in retrofit computer able to command the machine directly — Gravis's entire retrofit business model depends on that transition having happened. Second, an $8m UK government-backed CAM Pathfinder award for construction automation with Flannery, reported around August 2026, is a dated public-funding event that pulls demand forward. Not a 5 because neither is a hard mandate or a regulatory deadline with a fixed date — the joystick transition is gradual and undated in the evidence, and the Pathfinder grant is a funding programme rather than a customer mandate. The labour-shortage and infrastructure-boom arguments are the generic sector tailwind and carry no weight here. Sources: 1, 2, 3

Contradictions in the sources, left unresolved

- fundz.net's acquisitions tracker states SoftBank Group acquired Gravis Robotics on July 26, 2026, in a deal potentially exceeding a $500 million valuation. Every other financial-press source (ENR, Sifted, Forbes, The Robot Report, EU-Startups) describes the same capital event as a $200M Series A investment (not an acquisition) valuing the company at $1bn post-money.
- RocketReach lists 55 employees for Gravis Robotics, while Growjo states 26 employees plus a claimed $10.1M in annual revenue. Neither source is a primary company disclosure, and no company-confirmed headcount was found.
- Most press (TechFundingNews, EU-Startups, ENR, Forbes) names Ryan Luke Johns (CEO) and Dominic Jud (CTO) as co-founders with Marco Hutter as a board member/co-founder. A Chinese-language innovation listing (36kr) adds Marco Tranzatto and Burak Çizmeci as additional co-founders. CB Insights' people page states Gravis Robotics has 1 executive and that its founder is Marco Hutter, which does not corroborate the fuller founder list.

4. Humanoid — 59/100

United Kingdom · humanoid robots for industrial and manufacturing work

Criterion	Score	Weight	Points	Confidence	Hand-score
Field traction	2/5	30	12.0	HIGH	4/5 (H)
Team / execution	4/5	25	20.0	HIGH	4/5 (H)
Technology	3/5	20	12.0	HIGH	3/5 (M)
Market	3/5	15	9.0	HIGH	5/5 (H)
Timing	3/5	10	6.0	HIGH	3/5 (M)

Why these scores

Field traction. The confirmed activity is all proof-of-concept: a Siemens Erlangen trial (60 totes/hour), a six-week Ford Cologne trial (97% reliability, 83 picks/hour), and a Bosch intralogistics POC completed March 2026. Company evidence explicitly states no revenue-generating, non-POC deployment exists and that Beta robots only reach customer sites in Q4 2026. That would be a 1 on the anchor, but it is lifted above pure pilots/LOIs by Schaeffler's own press office confirming a *binding*, phased deployment and supply agreement for a four-digit unit count by 2032 with first systems live in Germany before end-2026 — a customer-side commitment stronger than an LOI, from a partner that also invested. It cannot reach 3 because there are no paying installs at 3-10 customers: the claimed 'nine industrial deployments with Fortune 500 customers' is single-sourced, self-reported and internally inconsistent (NVIDIA appears elsewhere only as a compute partner), and the 34,000 pre-orders/$2.4B is flagged unverified by multiple outlets. Sources: 1, 2, 3, 4, 5, 6, 7

Team / execution. Above a 3 because the senior bench is not merely credible-technical: CTO Jarad Cannon was CTO at Brain Corp through growth from 5 to over 40,000 deployed robots and spent six years at iRobot — i.e. someone who has shipped robots at fleet scale — and the founder previously scaled and sold a ~$1B-revenue manufacturing business, seeding this one with ~$30M of his own capital. Two hardware generations were unveiled inside 15 months (wheeled Alpha Sept 2025, bipedal Alpha Dec 2025) with shortening build cycles (7 months then 5 months), ~200 staff across London, Boston and Vancouver, and $270M raised including strategic money from Schaeffler and Bosch. It stops short of 5 because the generation timelines and the '48 hours to walking' claim are company-reported and explicitly flagged unverified by TheNextWeb, and neither generation has yet shipped in volume — the shipping-at-scale record belongs to a hired executive's prior employer, not to this team's own output. Sources: 1, 2, 3, 4, 5, 6

Technology. There is genuine engineering depth — a 41-DoF, 70kg wheeled platform with 15kg payload, and the KinetIQ four-layer stack with a trial-and-error learning variant and fleet orchestration — so this is above a 1. But the autonomy substrate is the standard purchasable one: Siemens' own release confirms Humanoid has integrated NVIDIA's full physical AI stack (Jetson Thor, Isaac Sim, Isaac Lab), which is available to every competitor. It falls short of 4-5 because nothing in the evidence compounds or is unbuyable: the actuators come from Schaeffler as external preferred supplier (>50% of demand), manufacturing is outsourced to Robert Bosch Robotics, KinetIQ's proprietary claim is company-attributed with no technical corroboration, and there is no certification moat or accumulating proprietary operating dataset described. Sources: 1, 2, 3, 4, 5

Market. Sizeable but demonstrably crowded and slow, which is exactly the 3 anchor. The founder himself names Tesla, Agility and Figure AI as competitors, and press notes Neura Robotics closed over $1B in Europe at a higher valuation — this is the most contested category in the sector. The cited market figure (~$38B by 2035, Goldman Sachs) is a future projection rather than an existing budget line, and the one binding commitment stretches procurement out to 2032, the kind of cycle that starves a startup. It is above a 1 because the targets — warehouse picking, kitting, machine feeding, load/unload at automotive and electronics plants — are large industrial segments where buyers like Schaeffler have signed for a RaaS structure, showing real willingness to spend; it does not reach 4-5 because no evidence shows buyers today making volume purchases in the humanoid category at scale. Sources: 1, 2, 3, 4, 5

Timing. The timing evidence is the sector-wide wave rather than a company-specific unlock. The strongest item — the roadmap being framed around the newly released NVIDIA Jetson Thor — is a component availability event equally available to every humanoid competitor, and it is a company-authored framing rather than an external mandate. The Siemens/NVIDIA CES partnership is a customer-side dateable event but it is Siemens' programme, not a mandate to buy from Humanoid. The remaining items ($18.8B robotics funding in 2026, $55.8B per Dealroom) are pure macro capital context, which is the definition of the default. Not a 1, because a concrete component release and a named customer AI-factory programme do anchor the 'why now'; not a 4-5, because no regulation, cost threshold crossing, or customer purchase mandate specific to this company is documented. Sources: 1, 2, 3, 4

Contradictions in the sources, left unresolved

- One trade outlet reports Humanoid claims nine industrial deployments with Fortune 500 customers including NVIDIA, SAP, and Siemens; this is not corroborated elsewhere, and NVIDIA's role is otherwise described as a compute/technology partner rather than a deployment customer, and no other source mentions an SAP deployment.
- Widely repeated company claim of 34,000 pre-orders worth $2.4 billion is explicitly flagged as self-reported and unverified by multiple outlets.
- Humanoid's claim to be Europe's first pure-play humanoid robotics unicorn is contested by TheNextWeb, which notes Germany's Neura Robotics closed a round exceeding $1 billion in June 2026 at a reportedly higher valuation.
- Company timelines (seven months for wheeled Alpha, five months for bipedal, 48 hours to stable walking) are self-reported and explicitly noted by TheNextWeb as unverified.

5. mimic robotics — 54/100

Switzerland · dexterous robotic manipulation using imitation learning

Criterion	Score	Weight	Points	Confidence	Hand-score
Field traction	2/5	30	12.0	HIGH	2/5 (H)
Team / execution	3/5	25	15.0	HIGH	4/5 (H)
Technology	3/5	20	12.0	HIGH	4/5 (M)
Market	3/5	15	9.0	MEDIUM	2/5 (L)
Timing	3/5	10	6.0	MEDIUM	3/5 (M)

Why these scores

Field traction. Every customer claim is unnamed and company-sourced: the November 2025 release asserts pilots with "Fortune 500 companies and global automotive brands" and logistics partners, and SiliconANGLE explicitly notes no names were disclosed. The ETH AI Center framing is "deepen pilot collaborations," i.e. still pilots. What lifts this above a pure 1 is one concrete, journalist-stated placement — FLUX-mimic taken "to the factory floor at Audi" in July 2026 — plus hiring for a Forward Deployed Robotics Engineer and Field Engineer, which implies real on-site work. But nothing shows a paying deployment, a unit count, or a repeat order, so it cannot reach the 3 anchor of paying deployments at ~3-10 customers; the only price point ($90k stations) is an unsourced Sacra estimate. Sources: 1, 2, 3, 4, 5, 6

Team / execution. This is the ETH spinout profile the anchor caps at 3: incorporated April 2024 out of the ETH Soft Robotics Lab, with Robert Katzschmann as scientific advisor and four founders drawn from the original research project — no evidence any of them has shipped hardware at scale before. Execution is credible and moving: pre-seed May 2024, a $16M seed in November 2025, the M1 hand and U1 exoskeleton launched, a peer-reviewable mimic-one paper, and the FLUX-mimic model with Black Forest Labs — essentially one shipped product generation plus a model release, not 2+ generations with shortening cycles. The About page claim of 50+ people with DeepMind and Tesla Optimus leadership would push toward 4, but it is company-stated, undated, and contradicted by press reports of 25 people at the seed, so it cannot carry the higher score. Sources: 1, 2, 3, 4, 5, 6, 7, 8

Technology. Real engineering depth: a tendon-driven M1 hand with 15 actuated DoF across 21 joints, forearm-mounted motors, and >25 kg cylindrical power grasp is not an off-the-shelf part, so this is above a 1. But the architecture is deliberately non-proprietary where it matters commercially — both the CTO and CPO say mimic buys widely-sold robot arms rather than building them — and the learning approach (diffusion policy for dexterous hand control) was published openly on arXiv, i.e. broadly available. The U1 passive exoskeleton for recording human hand demonstrations is the one element that could compound into a data flywheel, and FLUX-mimic is built with an external partner (Black Forest Labs) rather than on unbuyable internal assets; there is no evidence of proprietary data scale, an unbuyable component, or certification moat that a 5 requires. Sources: 1, 2, 3, 4, 5

Market. The target segments — automotive, manufacturing, logistics, retail — are large and already buy automation, and the German shortage of 5 million manufacturing workers by 2030 gives a real pain signal, which keeps this well above a 1 with no budget line. But the evidence also shows an unusually crowded field on both flanks: Sacra names Covariant (already earning subscription revenue), Skild AI ($300M) and FieldAI ($405M) as direct foundation-model manipulation comparables, Tracxn adds Agile Robots, Flexiv and Teradyne in dexterous industrial hardware, and Forbes situates mimic against Figure, Sanctuary and 1X. There is no evidence of a buyer today purchasing dexterous-hand stations at a specific budgeted line item — the only market sizing is a company-cited analyst projection to 2035 — so it does not reach the 5 anchor of urgent, already-budgeted spend. Sources: 1, 2, 3, 4, 5, 6

Timing. The timing case rests entirely on generic wave arguments: a projected 2030 German labor shortage (a slow demographic trend, not a dateable unlock), and mimic's own website framing that better models, data and cheap hardware are converging into a European robotics "gold rush" — a quote whose source and date the research could not even confirm. No regulation, no crossed component price threshold, no customer mandate is documented. The AWS Generative AI Accelerator selection and

Innosuisse grant are company milestones, not external market unlocks, so nothing here lifts it above the default; equally, the labor-shortage pressure is real enough that it is not a 1 with no external change at all. Sources: 1, 2, 3

Contradictions in the sources, left unresolved

- Press coverage of the November 2025 seed round consistently reports a 25-person team, while mimic's own About page (undated, referencing later product launches) claims a larger team of 50+ people; the two figures cannot be reconciled to a specific date from the sources found.
- The Tracxn company profile states mimic robotics was a seed company based in West Palm Beach (United States), founded in 2009, which contradicts every other source placing the company in Zurich and founded in 2024; this is treated as a data error rather than usable evidence.

4. What this rests on

Company	Sources retrieved	Evidence items kept	Searches	Budget spent?
ANYbotics	135	52	20/20	yes
Verity	133	47	20/20	yes
Gravis Robotics	81	52	20/20	yes
Humanoid	148	36	19/20	no
mimic robotics	125	38	20/20	yes

Every evidence item behind these scores carries a URL that was checked against the pages the research actually retrieved; a claim citing anything else was discarded before scoring. A `yes` in the last column means the research spent its whole search budget, so a gap in that company's evidence may be a limit of the search rather than an absence in the world.

The limitations of this method are stated in `00-scoping.md` §6. They are worth reading before the ranking is used for anything.

5. Provenance

- Collection and extraction: `claude-sonnet-5`
- Scoring: `claude-opus-5`
- Report assembly: no model. Every figure above is computed from the stored records, and every justification is quoted verbatim from the scoring step.
- Total API cost of the records behind this report: **$9.87**. This counts the files that survive on disk. Work that was discarded — a research run repeated at a larger search budget, an extraction rejected for unverifiable sources — is not included, so the amount actually spent building this was higher.

Regenerate with `.venv\Scripts\python.exe src\report.py`.

PART 2

The Thesis

Written 2026-08-18 on `claude-opus-5`, from the 5 companies scored in `03-market-map.md`.

The synthesis below is the model's. The criteria it argues on, and the weighting behind them, are set out in `00-scoping.md` and were fixed before any company was scored.

Executive summary

Across these five companies, nothing about the AI separated them. Field traction did, and traction tracked something unglamorous: whether the company had attached itself to a budget line and a compliance regime that already existed. ANYbotics (86) and Verity (85) sit at the top because their customers oONAAS, Outokumpu, Equinor, GE Vernova, DSV, KeHE, On — say in their own words that fleets are in daily service. The three below them are not technically weaker in any measurable way; they are earlier, and they are trying to create a purchase category rather than fill one. The most striking number in the set is that ANYbotics' single strongest defensive asset is an ATEX/IECEx Zone 1 certification, not a model, while Humanoid ($270M raised) and Gravis ($200M at a $1bn valuation) hold more capital than ANYbotics has raised cumulatively and cannot yet show a repeat order. On this evidence, European Physical AI is being won in the certification, channel and procurement layer, and the model layer is currently commodity. That is a claim about five companies, not a sector census — but it is the claim the evidence supports.

What the ranking shows

Technology did not sort anyone. Model-scored technology across all five: 4, 3, 3, 3, 3. Field traction across the same five: 5, 5, 3, 2, 2. The entire spread in the ranking comes from deployment, and almost none of it from engineering. That is not because the engineering is thin — a 41-DoF humanoid, a 15-DoF tendon-driven hand, a learning controller that infers soil state from hydraulic pressure are all real — but because under a rubric that asks *can a competitor buy this*, the answer was yes almost everywhere.

Nobody in this sample demonstrated a data flywheel. ANYbotics' Data Navigator processed 2,500+ inspections in two weeks, but only in pilot framing; Verity's scans are not shown to compound; mimic published its diffusion-policy approach on arXiv and buys its arms; Humanoid runs NVIDIA's Isaac/Jetson Thor stack, available to every competitor, with actuators from Schaeffler and manufacturing outsourced to Bosch; Gravis' CEO volunteers that the OEM switch to electronic joysticks makes retrofit computers easy to plug in — which opens the same door for SafeAI and Hive Autonomy. The one durable barrier found anywhere in the set is regulatory: ANYbotics' ATEX/IECEx Zone 1 IIB certification, a path that ran 2019–2026.

Capital is decoupled from deployment, and the sign is inverted at the top. Ranked by money: Humanoid ($270M), Gravis ($200M Series A at $1bn post-money), ANYbotics (~$150M cumulative), Verity (~$50M+ across tranches), mimic ($16M seed). Ranked by traction: ANYbotics and Verity at 5, Gravis 3, Humanoid and mimic 2. The two companies with confirmed multi-site paying fleets have raised less than the two with none.

The discriminating evidence class is the customer's own newsroom. Every unit count in this map traces back to the vendor — ANYbotics' ~200 units, Verity's 150 sites, Gravis' \"seven countries,\" Humanoid's 34,000 pre-orders, mimic's unnamed Fortune 500 pilots. What separated the top two was that third parties published on their own letterhead: Outokumpu naming three robots across three plants with a repeat order, PETRONAS confirming a signed 2022 commercial agreement, GE Vernova folding ANYmal into its own APM software, DSV's operations manager framing drones against contractual SLAs. Humanoid has exactly one such artefact — Schaeffler's binding four-digit agreement — and it is the reason it scores 2 rather than 1.

ETH pedigree is table stakes; shipped-at-scale experience is the lift. Four of five are ETH spin-offs and the rubric caps pedigree at 3. What pushed teams above it was concrete: Raffaello D'Andrea having co-founded Kiva (Verity, 5/5), ANYbotics' four hardware generations, Jarad Cannon's Brain Corp record at Humanoid (4/5). mimic, with the same lab lineage and one product generation, sits at 3.

Winners attach; laggards create. ANYbotics sells into asset-integrity budgets with SLB as channel; Verity sells into inventory-accuracy SLAs; Gravis sells onto plant-hire fleet budgets at Flannery and Holcim. Humanoid and mimic sell general-purpose labour — a line item that does not exist yet, with a Goldman projection to 2035 and a Schaeffler ramp to 2032 standing in for it.

Calibration reads as systematic conservatism on the interpretive criteria. The model scored below the hand on technology (−5 notches) and market (−4), and slightly above on timing (+1). Rank order was identical on both sides. The ordering is robust; the absolute scores are a floor, and the 86-vs-85 gap at the top is noise.

The thesis

European Physical AI is not currently a competition between learning systems. It is a competition for procurement slots, and the moats being built are certifications, integrations and channel relationships — assets that look boring on a pitch deck and cannot be trained.

The strongest single fact in this entire map is that the most defensible thing ANYbotics owns is a hazardous-area certification, not a policy network. It has only ~8 patents filed and paid Boston Dynamics a licensing fee in 2019; its actuator is now made by maxon. What competitors cannot shortcut is the seven-year ATEX/IECEx Zone 1 path, and what makes its pipeline addressable is that GE Vernova embedded the robot inside its own APM software and SLB made it a preferred ground robotics supplier. Verity's second place rests on the same structure minus the certification: purpose-built airframes, yes, but its score is carried by DSV framing the drones against contractual SLAs and KeHE rolling from one Arizona site to a hundredth installation. In both cases the defensible asset sits between the robot and the customer's existing spend, not inside the robot.

From this I take a position that can be wrong: **the capital currently flowing to general-purpose embodiment in Europe is mispriced relative to the capital flowing to certified, budget-attached single-purpose autonomy, and the next 24 months will show it.** Humanoid has raised $270M and Gravis carries a $1bn post-money valuation; between them the evidence shows zero repeat orders and zero confirmed non-POC paying deployments. Their bottlenecks, on this evidence, are not model quality — Humanoid already runs the same NVIDIA stack everyone else can buy, and hit 97% reliability at Ford Cologne over six weeks. The bottlenecks are that Schaeffler's binding commitment ramps to 2032, that the actuators and the factory both belong to other companies, and that the founder himself names Tesla, Agility and Figure as competitors while Neura closed over $1B in Germany. A 2032 ramp is not a market; it is an option on one.

The corollary for Switzerland specifically: the ETH/EPFL pipeline is producing world-class perception-and-control teams at a rate that has made technical competence non-differentiating within this sample. Four of five companies come from the same two institutions and cluster at 3–4 on technology. If that is right, the scarce Swiss resource is not another lab spin-out — it is people who have shipped hardware fleets and know how to get a robot onto a Zone 1 site or inside an OEM's software. Verity bought that with D'Andrea's Kiva history; Humanoid hired it in Jarad Cannon; mimic, on public evidence, has neither, and it ranks last.

What follows practically: value the certification and the customer's own press release, discount the pre-order book and the TAM slide. On this map, an announcement from Outokumpu's newsroom was worth more than $2.4B of self-reported pre-orders.

What would falsify it

The single strongest falsifier is Humanoid's Schaeffler agreement converting on schedule. Schaeffler's own press office committed to first systems live in Germany before end-2026, with Beta robots reaching customer sites in Q4 2026, ramping to a four-digit unit count by 2032. If by mid-2027 Schaeffler, Bosch or Ford publish, on their own letterhead, named multi-site paying (non-POC) fleets — with unit counts — then general-purpose humanoids will have crossed from pilot to procurement roughly five years faster than the \"2032 ramp is an option, not a market\" claim assumes, and the capital allocation I called mispriced will look correctly early.

Gravis' capital-ahead-of-traction position paying off. Gravis debuted its second-generation Copilot at CONEXPO in March 2026 following the $200M SoftBank round. If by end-2027 Flannery Plant Hire, Holcim or Kibag announce fleet-scale retrofit orders with disclosed unit counts and repeat purchases — rather than another \"UK-first trial\" at a Manchester Airport-type site — then traction lagged capital by design and the ordering logic here (traction leads, money follows) is too slow a read.

The certification moat proving shortcuttable. If Boston Dynamics, Unitree or a Chinese entrant announces ATEX/IECEx Zone 1 certification for a legged platform before ANYmal X's full commercial launch, or if ANYbotics slips that 2026 launch again (it has already moved from a 2022 unveiling), the specific thing I identified as the sector's only real barrier is either copyable or too slow to monetise — and the thesis loses its anchor case.

Compliance cutting the other way. Verity's FCC Covered List approval expires 31 December 2026. If it is not renewed and Verity loses US market access, then the \"regulation as moat\" reading needs replacing with \"regulation as random shock,\" and a certified position is not an asset but a liability with a renewal date.

A demonstrated flywheel. If any company here publishes third-party-verifiable evidence that accumulated field data measurably improved model performance in a way competitors cannot replicate — ANYbotics' Data Navigator moving from pilot description to audited fleet-level performance gains, or mimic showing that U1 exoskeleton demonstration data yields policies FLUX-mimic competitors cannot match — then the model layer is not commodity and the ranking's technology scores were measuring the wrong thing.

Displacement at the top by a well-funded generalist. If Neura Robotics (>$1B raised, Germany) reaches confirmed multi-site paying fleets in industrial settings before ANYbotics or Verity expand theirs, then raising big and going broad beats attaching narrowly, and the ordering here inverts.

An account loss. If Gather AI (post-Ware Robotics acquisition) or Corvus displaces Verity at a named account — IKEA/Ingka, DSV, KeHE — then installed base without a technology moat is not sticky, which is the load-bearing assumption in ranking Verity second on a technology score of 3.

Where this method is weakest

Five companies, chosen to test the scoring method, cannot describe a sector. Four are Swiss, one British. Three sit in inspection, warehousing and construction. There is no agriculture, surgical, defence, last-mile, or fixed-infrastructure AMR company in the set — which means the largest deployed Physical AI businesses in Europe by units and revenue are simply absent. An Exotec- or AutoStore-class company would almost certainly outrank ANYbotics on field traction, and Neura Robotics appears here only as a contradiction inside Humanoid's profile. Any statement about \"European Physical AI\" in this document is really a statement about these five and should be read that way.

The ranking reads public disclosure, not operations. One search pass, no customer references called, no revenue, gross margin, churn, renewal rate, RaaS contract length or service cost visible for anyone. That matters most where it hurts the top: ANYbotics' and Verity's traction scores rest on deployments that are *announced*, and we cannot see what happens after. The only dead pilot in the whole map — Verity's original Maersk site in Miami — surfaced because a single outlet, Supply Chain Dive, went and asked. The base rate of quietly dead installations is invisible by construction, and it is the number that would most change the ranking.

Traction as measured rewards customers who publish. ANYbotics scores 5 partly because Outokumpu, PETRONAS, Equinor and GE Vernova run press offices that name suppliers. A company selling equally real fleets to buyers under NDA — common in defence, pharma, semiconductors — would score 2 or 3 on identical underlying reality. This is a systematic bias, not noise, and it favours industrials-facing vendors over anyone selling into secretive procurement.

Field traction at 30% makes the ranking a snapshot of the present, not a forecast. mimic was incorporated in April 2024; it cannot score well on units deployed and its 54 is not a prediction that it fails. The rubric measures deployed reality, which is the right thing to measure for a market map and the wrong thing to confuse with expected value. Readers looking for who wins in 2030 are reading the wrong instrument.

The double-count and the timing cap. ANYbotics' ATEX certification earns credit twice — once as technology moat, once as dated timing unlock — which inflates the single most important fact in the set. Meanwhile timing is capped at 10 by design, so a company holding a genuine external regulatory deadline could gain at most 10 points from it; if such a mandate exists in any of these markets, this method structurally under-weights it.

The underlying record is unaudited and internally contradictory. Even the best-evidenced company has no reconciled ledger: ANYbotics' headcount ranges 100–260 across sources and its funding total is variously $130M, $150M and €127M; Gravis' headcount is 26 or 55, and one tracker states SoftBank *acquired* the company while every financial outlet describes a minority investment; Tracxn places mimic in West Palm Beach, founded 2009. Confidence flags of HIGH in these profiles mean \"the evidence class is strong,\" not \"the numbers are verified.\"

Calibration bounds what the numbers mean. Twelve of 25 criterion scores matched the blind hand-score, and rank order matched exactly — so the ordering survives the disagreement. But the model ran −5 notches on technology and −4 on market against the human, meaning the two most interpretive criteria are where reasonable scorers diverge most, and always in the direction of the model being stricter. Treat absolute scores as a floor. Treat 86 versus 85 as a tie. And note that a scorer who credited engineering depth and TAM narrative the way the hand-scorer did would compress this whole field — hand scores run 93, 92, 85, 77, 60 — into a much narrower band, which would weaken, though not reverse, the argument that traction is what separates these companies.

The limitations of the method are stated at greater length in `00-scoping.md` §6.

PART 3

Method and Limitations

This document is the actual deliverable of the project. Everything downstream is execution. Nothing gets coded until every section below is filled in and defensible out loud.

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1. Sub-sector definition

Sector: Physical AI. Companies whose core product is a machine that perceives and acts in the physical world, and whose value comes primarily from onboard autonomy rather than from the mechanics. Mobile robots, drones, autonomous industrial machines, manipulation, humanoids.

Geography: Switzerland and Europe (EU + UK + Norway), by headquarters.

In scope — a company qualifies if all four hold: - Its main commercial product is a physical system, and the differentiator is the AI on board - It is independently financed — a startup or scale-up, not a division of a listed group - Headquarters in Switzerland or Europe - At least one disclosed funding round, or public evidence of revenue

Out of scope — explicitly excluded, with the reason: - **Pure software, simulation or foundation-model companies** — no physical product, so the same criteria would not measure the same thing - **Industrial automation incumbents** (ABB, KUKA, Stäubli) — mature and orders of magnitude larger; including them compresses every startup into the bottom of the scale - **Swiss and European R&D sites of foreign tech groups** (Google Zurich, Apple, NVIDIA, Meta, Disney Research) — scoring Alphabet on "amount raised" is meaningless. Listed unscored in the annex, section 7 - **Academic labs and unincorporated spin-off projects** — no company to score yet

Edge case rulings, decided in advance: - *Founded in Europe, HQ later moved to the US* → out of the main ranking, listed in the annex and flagged. The migration is itself a finding worth discussing in the thesis. - *European HQ, manufacturing in Asia* → in scope. Where it builds does not change what it is. - *Acquired by a larger group but still operating as a distinct product line* → out of the ranking, noted in the annex. Its funding trajectory stopped being a signal on acquisition.

> *Why the boundary matters: an interviewer will test it with an edge case. This definition is built to survive "what about company X?" — every exclusion above has a stated reason, and the reason is always the same one: the criteria must measure comparable things.*

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2. Company list

26 companies. Funding figures are deliberately absent here — collecting and sourcing them is the job of steps 1 and 2. This list is names, geography and segment only.

Sources used to build it: the Swiss Robotics Startups ecosystem map (~100 entries, filtered against the section 1 criteria), Greater Zurich Area's robotics selection, Top100 Startups Switzerland, and EU-Startups / Tech.eu funding coverage from 2025–2026.

#	Company	Country	Sub-segment	How I found it
1	ANYbotics	CH — Zurich	Legged robots, industrial inspection	Known ETH spin-off
2	Verity	CH — Zurich	Autonomous warehouse inventory drones	Ecosystem map
3	Voliro	CH — Zurich	Contact-inspection drones	Ecosystem map
4	Flyability	CH — Lausanne	Confined-space inspection drones	Known EPFL spin-off
5	Wingtra	CH — Zurich	VTOL mapping and survey drones	Ecosystem map
6	mimic robotics	CH — Zurich	Dexterous manipulation	EU-Startups, Nov 2025
7	Ascento	CH — Zurich	Wheeled-legged security patrol robots	Ecosystem map
8	Gravis Robotics	CH — Zurich	Autonomous heavy construction machinery	Greater Zurich selection
9	LOXO	CH — Bern	Autonomous delivery vehicles	Ecosystem map
10	Ecorobotix	CH — Yverdon	Precision agriculture, smart spraying	Ecosystem map
11	Tethys Robotics	CH — Zurich	Autonomous underwater robots	Greater Zurich selection
12	Saeki	CH — Zurich	Robotic fabrication for construction	Ecosystem map
13	NEURA Robotics	DE — Metzingen	Cognitive robots and humanoids	Largest EU round, 2026
14	Agile Robots	DE — Munich	Force-controlled manipulation	Sector coverage
15	Sereact	DE — Stuttgart	Embodied AI for robotic picking	EU funding coverage 2026
16	Robco	DE — Munich	Modular industrial robots for SMEs	Sector coverage
17	Fernride	DE — Munich	Autonomous yard and port logistics	Sector coverage
18	Humanoid	UK — London	Industrial humanoid robots	Tech.eu, Jul 2026
19	Dexory	UK — Oxford	Warehouse inventory robots	Sector coverage
20	Wayve	UK — London	Embodied AI for driving — see edge case	Sector coverage
21	Exotec	FR — Lille	Warehouse goods-to-person robotics	Sector coverage
22	Donecle	FR — Toulouse	Automated aircraft inspection drones	EU-Startups, Apr 2026
23	AgreenCulture	FR — Toulouse	Autonomous agricultural robots	Sector coverage
24	PAL Robotics	ES — Barcelona	Humanoid and service robots	Sector coverage
25	THEKER	ES — Barcelona	Humanoid manipulation	EU funding coverage 2026
26	Oversonic Robotics	IT — Besana Brianza	Humanoid robots for industry	Sector coverage

2b. Edge cases — ruled on before scoring

Applying section 1 to the grey zone. Each ruling is stated so it can be defended.

Company	Ruling	Reason
Auterion (CH)	**Out**	Drone autonomy software running on third-party airframes. No own physical product.
Energy Robotics (DE)	**Out**	Inspection software layered on other vendors' robots. Same reason as Auterion.
Bota Systems (CH)	**Out**	Force-torque sensor supplier — a component, not a system that perceives and acts.
Distalmotion (CH)	**Out**	Surgical robotics. Strong Swiss player, but medtech regulation and sales cycles make it incomparable to industrial Physical AI.
Wayve (UK)	**In, flagged**	Does not build the vehicle, which strains the "physical product" test. Kept because the autonomy stack *is* the product and it is deployed on real roads. Reversible if the flag proves distorting.

> *The Wayve ruling is the one to expect a question on. The honest answer is that it sits on the boundary, the boundary was drawn before the company was considered, and the decision is documented rather than silent.*

3. Scoring criteria and weights

Weights must sum to 100. The justification column is what gets discussed in an interview — it is more important than the number itself.

Criterion	Weight	What it measures	Why this weight
Field traction	**30**	Units deployed at paying industrial customers. Funding is a secondary indicator only.	In Physical AI the binding constraint is deployment, not modelling. Anyone can demo; almost nobody survives contact with a real site.
Team / execution	**25**	Founder background, hardware iteration speed, ability to hire scarce robotics talent	Hardware punishes slow teams. Iteration speed is the single best predictor of who is still standing in three years.
Technology / product	**20**	Maturity, technical moat, defensibility	Weighted below traction on purpose: robotics moats erode fast because everyone reads the same papers. What defends is accumulated real-world data, which shows up under traction.
Market	**15**	Addressable size, segment growth rate	Discriminates weakly here — most of the panel targets comparable industrial segments, so it separates few companies.
Timing	**10**	Why now — what changed that makes this possible today	Deliberately low. See section 3b.
Total	**100**		

3b. Two positions this weighting takes

Traction is not the amount raised. It is redefined as units deployed at paying industrial customers, with funding demoted to a secondary indicator. Dollars raised is a lagging, noisy signal: a €100M round reports what a group of investors believed eighteen months ago. Robots running on a customer site report what works now.

Timing is capped at 10 on purpose. The "why now" of Physical AI — cheap sensors, vision- language-action models, industrial labour shortage — is identical for every company in the panel. A criterion that awards everyone the same score does not rank anything; it only inflates the totals. Keeping it at 10 acknowledges that it matters to the sector without pretending it separates companies within it.

4. Scoring scale

A 1–5 scale with written anchors for every criterion, so that two runs of the pipeline produce comparable scores. Without anchors the model drifts between runs and the ranking means nothing.

Scores 2 and 4 are intermediate positions between the anchors below.

Field traction — weight 30

Score	Anchor
1	Demos, pilots and letters of intent only. No customer paying for a deployment, or deployments only at investor and partner sites.
3	Paying deployments at roughly 3–10 customers, still sold project by project, each installation substantially custom.
5	Repeat orders from several industrial customers, tens of units in routine daily service, and at least one customer that has moved past an initial pilot into a fleet.

> *The discriminator is the* ***pilot-to-fleet transition****, not the number of pilots. Robotics is full of companies with fifty pilots and no second order — that is "pilot purgatory", and it scores a 2, not a 4. A single customer that scaled from three units to thirty is worth more evidence than twenty logos on a website.*

Team / execution — weight 25

Score	Anchor
1	First-time founders with no record of shipping hardware, no visible iteration, flat or shrinking team.
3	Credible technical founders with a strong research pedigree, one product generation shipped, steady hiring.
5	Founders who have shipped hardware at scale before, or two-plus product generations shipped at this company with visibly shortening cycles, and the pull to hire senior robotics people away from big tech.

> ***An ETH or EPFL pedigree caps at 3 in this panel.*** *Nearly every Swiss company on the list has one, so it is table stakes, not a differentiator. Anything that everyone shares cannot separate anyone — the same logic that caps the timing criterion at 10.*

Technology / product — weight 20

Score	Anchor
1	The capability is demonstrated in published research by others and the demo could be reproduced with off-the-shelf components.
3	Real engineering depth — hard integration work, reliability in the field — but the underlying approach is broadly available to competitors.
5	An advantage that compounds: a proprietary real-world data flywheel, a hardware component competitors cannot buy, or a certification that takes years to replicate.

> *A 5 requires something that **gets stronger with use**. A clever architecture does not qualify; everyone reads the same papers. This is the criterion where the weighting argument from section 3b cashes out.*

Market — weight 15

Score	Anchor
1	Narrow niche with few buyers, or a segment where no budget line exists for this category of spend.
3	Genuinely sizeable segment, but crowded, or with procurement cycles long enough to starve a startup.
5	Large segment with an urgent and already-budgeted pain — labour shortage, regulatory inspection mandate — where the buyer purchases in this category today.

> *Judge the **existing budget line**, not the headline market size. A TAM is easy to inflate and nobody can check it. A buyer who already has a line item for inspection is a buyer who can sign.*

Timing — weight 10

Score	Anchor
1	Could have been built five years ago and was not, with no external change explaining why now.
3	Rides the general Physical AI wave — the same "why now" as every other company in the panel. **This is the default score here.**
5	A specific, dateable unlock the company is positioned for: a regulation entering into force, a component crossing a price threshold, a customer-side mandate.

> *Because the generic wave scores a 3 for everyone, only a specific unlock earns a 5. This is what keeps a low-weight criterion from quietly inflating every total.*

4b. Handling missing evidence

The pipeline will not find complete data for all 26 companies. Younger and less publicised companies will have thinner public footprints, and that must not be silently rewarded or punished by guesswork.

Rule: where evidence for a criterion is absent, the score is capped at 2 and flagged as low confidence. The model must never infer a score from what is plausible for a company of that type.

Absence of evidence is recorded as absence, not averaged away into a middling score. Every low-confidence flag is listed in the final report so a reader can see exactly which parts of the ranking rest on thin data.

4c. Sourcing the traction criterion

There is a real tension in this design, and it is better stated than discovered: field traction carries the highest weight (30) and is the hardest evidence to obtain publicly. Industrial buyers rarely disclose deployment volumes. Left alone, the most important criterion would carry the most low-confidence flags.

The weight is not lowered — the argument for it holds. Instead, step 1 searches for traction evidence deliberately and beyond the company's own marketing:

Source	What it reveals
Customer-side press releases	The buyer announcing a rollout is stronger evidence than the vendor announcing a win
Published case studies	Usually name a site and sometimes a unit count
Job postings	A company hiring ten field deployment engineers has deployments. Hiring only researchers does not.
Trade and industry press	Covers installations that never reach the startup press
Regulatory and tender records	Public-sector deployments leave a paper trail

Job postings are an **indirect indicator** and are labelled as such in the extracted data. They support a score, they never establish one on their own.

4d. Model allocation across the pipeline

Two models, assigned per step rather than one model for the whole pipeline. The principle: **spend capability where judgment happens, not uniformly.**

Step	Model	Reasoning
1. Source collection	`claude-sonnet-5`	High input volume (search results), low judgment. Finding and returning pages is not where model capability separates.
2. Structured extraction	`claude-sonnet-5`	Reading a page and filling a schema field with its source URL. Mechanical once the schema is fixed.
3. Weighted scoring	`claude-opus-5`	The only step that is pure judgment: arbitrating between the 1/3/5 anchors, applying the missing-evidence cap, and writing a justification that survives challenge. This is the step the project is actually about.
4. Report generation	**none**	Formatting already-decided content — and once that is true, a model is the wrong tool. Assembled in Python from the stored records, so no figure can drift from the record it came from. Free and reproducible. Sonnet was allocated here originally; the change is recorded rather than silent.
5. Thesis and executive summary	`claude-opus-5`	Synthesis across 26 scored companies — pattern-finding, not transcription.

Cost at August 2026 list prices: Opus is 2.5× Sonnet per token on both input and output. Running the whole pipeline on Opus would roughly double the project cost for a quality gain confined to steps 3 and 5.

Consequence to respect on any rerun: step 3 stays on Opus. A scoring pass run on a different model is not comparable to the existing one, and the calibration in §5 would have to be redone.

5. Calibration set

Five companies scored **by hand, before running the pipeline**. The gap between these scores and the model's output is measured and discussed in the final report.

This is the section that makes the project credible. Do not skip it.

Hand-scoring completed 2026-08-18, blind, before any pipeline code was written. Full per-criterion scores, confidence levels, and justifications are in `calibration-worksheet.md`.

Scored 2026-08-18 on `claude-opus-5`. The model saw only the extracted evidence: never the open web, never these hand-scores, and never another company. It did not compute its own total — the weighting is applied in `src/scoring.py`, by the same arithmetic as the worksheet.

Company	My score	Model score	Gap	Who was right, and why
ANYbotics	93/100	86/100	−7	**Model, narrowly, on technology.** I scored the moat 5; it argued 4 by conceding the ATEX/IECEx Zone 1 certification and then pricing the IP against it — 8 patents filed, a 2019 patent licence paid to Boston Dynamics, and the ANYdrive actuator now built by an outside supplier. That is a specific case I did not make. Traction, team and timing agree exactly.
Verity	92/100	85/100	−7	**Draw, same shape as ANYbotics.** Identical pattern: agreement on traction and team, one notch down on technology and market. Two companies I rated within a point of each other are still within a point of each other.
Gravis Robotics	85/100	70/100	−15	**Model, on technology (−2).** I scored 5. It found no compounding advantage: commodity cameras, lidar and GNSS, no certification barrier, BuiltWorlds placing Gravis beside Hive Autonomy and SafeAI doing the same retrofit approach, and the CEO's own point that electronic joystick signals make retrofits easy — which opens the same door to competitors. I scored this the day after a $200M round; I should treat that as a possible influence on my read.
Humanoid	77/100	59/100	−18	**Model, and the test inverted — see below.** Traction −2 and market −2 carry the whole gap.
mimic robotics	60/100	54/100	−6	**Draw, and the closest agreement in the set.** Traction identical at 2. It is +1 on market where I was harsh, −1 on technology and team. Nothing here separates us.

Agreement across the 25 criterion scores: **12 exact, 22 within one notch**, three disagreements of two notches (Gravis technology, Humanoid traction, Humanoid market).

The model is stricter than me on every company, never more generous. A directional bias is a more useful finding than scatter would have been: it is one correction to argue about, not five.

The ranking is identical. ANYbotics, Verity, Gravis Robotics, Humanoid, mimic robotics, in that order, on both sides. Where the totals move, the order does not.

Where the disagreement actually sits, summed across the five companies in notches:

Criterion	Sum of gaps	Reading
Team / execution	−1	Near-consensus. We read founders and hiring the same way.
Timing	+1	Consensus, helped by the 3 default doing its job.
Field traction	−3	Concentrated: Humanoid −2 and Gravis −1. The other three agree exactly.
Market	−4	**The model discriminates where I did not.** I gave four 5s (§5, third observation, called this out as a flaw in my own scoring). It gave 4/4/4/3/3.
Technology	−5	**The real disagreement.** It reads moats as thinner than I do, on four of five companies.

Technology is the argument to prepare. My scores lean on engineering depth; its scores ask what *compounds* — which is closer to what the anchor actually says. That is worth conceding or defending explicitly, not splitting.

Three observations from the hand-scoring pass, recorded before seeing any model output:

- **The Humanoid test no longer isolates what it was designed to test.** It was chosen to check whether the model conflates funding with traction — traction was expected to score low. It scored 4, on the strength of a dated, binding Schaeffler contract. The reasoning holds, but if the model also scores it high, the two possible causes (reading the contract vs. being swayed by headline numbers) become indistinguishable.

Resolved by the scored run, in the opposite direction to the one the test anticipated. The model scored Humanoid's traction **2**, against my 4 — it was the stricter reader, not the looser one. It credits the Schaeffler agreement for lifting the score above a pure 1, then refuses 3 because there are no paying installs at three to ten customers, and it flags the "nine industrial deployments with Fortune 500 customers" as single-sourced, self-reported and internally inconsistent, and the 34,000 pre-orders as unverified. On the evidence in front of it, that reading of the anchor is better than mine: I let a binding future commitment stand in for present deployment. This one disagreement is −12 of the −18 point gap on Humanoid. - **Gravis Robotics no longer tests the coverage floor.** It was chosen for near-zero public coverage. A $200M SoftBank round announced 2026-08-17 changed that overnight; it scored 85 with HIGH confidence on four of five criteria. **mimic robotics is now the only functioning missing-evidence test** — it produced the single LOW-confidence flag in the set. - **Market does not discriminate.** Four of five companies scored 5. That is the same failure the weighting section identifies for Timing, and it suggests the Market "5" anchor is too easy to reach. Weight is only 15, so the effect on the ranking is bounded — but it is a real finding about the criteria, not about the companies.

5b. Observations from the extraction pass, before any scoring

Step 2 ran on all five companies on 2026-08-18. Two findings are recorded here, before the scores exist, because they change how the table above should be read.

- **The missing-evidence test did not fire on the company it was built for.** mimic robotics is the only company in the set still carrying a thin public footprint, and it was hand-scored 60/100 largely for that reason. Extraction returned `field_traction: found` at HIGH confidence. Its step 1 report states plainly that no named paying customer was confirmed by any third party. Neither `searched_not_found` nor `not_searched` was assigned, so the §4b cap never engaged.

- **A single item carried that rating, and it is not traction evidence.** HIGH requires one direct claim from an independent source. For mimic robotics that claim is a July 2026 trade article reporting a new video-action model developed with Black Forest Labs — a product announcement naming no customer, site, unit count, or agreement. The other six items are all `indirect`. The same count is 14 for Verity, 9 for ANYbotics, 8 for Gravis Robotics and 4 for Humanoid. All five are rated HIGH.

Corrected after scoring — the warning above applies to step 2 only. Written before step 3 ran, this section concluded that the pipeline had not measured mimic robotics' weakness. The scored result contradicts that. Opus rated its field traction **2/5, the same value I gave it by hand**, and its justification names precisely what the confidence flag missed: *every customer claim is unnamed and company-sourced*, SiliconANGLE *explicitly notes no names were disclosed*, and nothing shows *a paying deployment, a unit count, or a repeat order*.

So the gap is real but it is contained. Step 2's confidence rule was fooled; step 3 was not, because it reads `evidence_grade` and `attributed_to` rather than the summary flag. The honest statement is narrower than the one first recorded here: **a HIGH confidence flag is not a statement about how strong the evidence is, and should not be read as one** — in the report, or by anyone using this pipeline. The scores themselves are not affected.

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6. Known limitations

Stated upfront, not discovered by the interviewer.

A search budget manufactures absence. The first ANYbotics collection ran with a budget of 12 searches and concluded that no customer-side press release, tender record, or regulatory filing named the company. Re-run at 20 with the same prompt, it found four — Outokumpu, PETRONAS, Equinor, GE Vernova — plus patent data it had previously reported as unsearched. Under §4b that difference is not cosmetic: absent evidence caps a score at 2, so **the search budget was an input to the ranking, not only to the cost.** Two mitigations are in place: the budget is now 20, and step 2 records `searched_not_found` separately from `not_searched` so that only genuine absence triggers the cap. Neither proves sufficiency — 20 searches are still fully consumed on data-rich companies, so the ceiling may still bind on the largest ones. Every capped score should be read as *not found within 20 searches*, never as *does not exist*.

Confidence measures the kind of source, not the weight of the evidence. It is computed in Python, not asked of the model: HIGH requires at least one direct claim from an independent source (customer-side, tender record, or trade press), MEDIUM is everything else with evidence, LOW is absence. This makes §4b enforceable and removes run-to-run drift, at a stated cost — one independent source and six both return HIGH. Confidence here answers *who said it*, not *how much of it there is*.

Narrowed after the first run. HIGH now requires two independent direct sources rather than one, so it means *corroborated*. Seven of the flags across the five companies moved down as a result. The limitation is real but smaller than first written: above two, volume still does not move the flag.

Nothing checked that a piece of evidence belonged to the criterion it was filed under. Extraction assigns each item to a criterion, and the confidence rule then counts items; neither step asks whether the claim is *about* that criterion. Measured on mimic robotics (§5b): a product announcement filed under field traction was graded a direct claim from an independent source and, on its own, lifted the company's weakest criterion to HIGH. Combined with the binary rule above — one qualifying item scores the same as fourteen — a single misfiled item is enough to erase a coverage gap. Reading the evidence list is part of reading the score here, not an optional audit on top of it.

Addressed, and the fix is not free of its own assumption. An evidence item must now name what makes it direct — a named customer, a unit count, a signed agreement, a dated deployment, or a regulatory record — and one that names none of those is recorded as indirect regardless of the grade the model assigned. The scoring prompt also now states that one fact scores under one criterion only, with the ATEX case named. Both rules take effect on the next run of their step; the scores in §5 predate them. The assumption they rest on is that the model labels its own reasoning honestly, which is weaker than a check that reads the claim itself.

Public evidence only. No interviews, no customer references, no proprietary databases. This favours companies that publicise well over companies that deploy well — exactly the confusion §3b sets out to avoid. Searching customer-side sources rather than vendor announcements (§4c) reduces the bias; it does not remove it. The step 1 prompt also does not explicitly request German-, French-, or Italian-language sources, which matters for a panel centred on Switzerland. Whether that changes coverage materially has not been measured.

Errors in step 1 are invisible to step 3. The scoring model never sees the web — it scores the evidence document that collection produced. A source that was missed, or misread, propagates silently into the ranking with no mechanism downstream to catch it. The audit trail exists (every claim carries its URL) but auditing it is a manual act, not an automated check.

The ranking is a snapshot, and the panel moves faster than the snapshot. Gravis Robotics went from near-zero public coverage to a $200M SoftBank round announced 2026-08-17 — during the scoping work itself. Any score in this report is a statement about the evidence available on its collection date, which is recorded per company.

The calibration set is five companies scored by one person. It measures agreement between a hand-scorer and the model, not correctness of either. Two of the five have also lost the property they were chosen to test: Humanoid no longer isolates funding-versus-traction, and Gravis no longer tests the coverage floor (§5). mimic robotics is the only functioning missing-evidence test left, so the missing-evidence machinery rests on a sample of one.

A quarter of the weight sits on criteria that barely separate the panel. Timing is capped at 10 because the "why now" is identical sector-wide (§3b) — that is deliberate. Market at 15 was expected to discriminate and, on the calibration set, did not: four of five companies scored 5. Together that is 25 points of 100 doing little ranking work, which concentrates the real discrimination in traction and team. Raising the Market bar is a change to consider after the full run, not mid-flight.

One fact scores twice. ANYbotics' ATEX/IECEx Zone 1 certification is credited under technology, as the multi-year certification moat the 5 anchor names, and again under timing, as a dated external unlock. It is the single most important fact in the set, and the rubric counts it in two places, which inflates the company that holds it. Found by the pipeline's own synthesis step rather than by me, which is the reason it is written here instead of quietly patched: a criterion set that can double-count is a finding about the criteria. The fix is to decide which criterion owns a certification before the next run, not to adjust one score after the fact.

7. Annex — ecosystem players, not scored

Excluded from the ranking because the scoring criteria do not apply to them, but they shape the sector and belong in the picture.

Foreign corporate R&D sites in Switzerland

Not scored — the criteria are built for independently financed companies. Listed because they compete for the same talent pool as every company in the ranking, and because they are where a share of Swiss robotics and computer-vision work actually happens.

Group	Site	Relevance to Physical AI
Google / DeepMind	Zurich	Largest Google R&D centre outside the US; robotics and computer vision
Amazon	Zurich	Reinforced by the RIVR acquisition, see below
Apple	Zurich	AI and computer vision lab
Meta	Zurich	Computer vision
Microsoft	Zurich	Mixed reality and AI
NVIDIA	Zurich	Robotics and simulation
Disney Research	Zurich	Character robotics, long-standing ETH partnership
IBM Research	Rüschlikon	Long-standing European research site
ABB	Zurich region	Not foreign, but a listed incumbent — excluded from the ranking for scale

To verify and complete in step 1 — headcount and current robotics activity per site.

Acquired companies

The most interesting rows in this document. Both were ETH-originated, both were acquired before reaching independent scale.

Company	Acquirer	Year	Signal
RIVR (ex Swiss-Mile)	Amazon (US)	March 2026	Wheeled quadrupeds for last-mile delivery. Raised ~$25M incl. Bezos Expeditions before exit.
Sevensense Robotics	ABB (CH/SE)	2024	Visual navigation for mobile robots. Absorbed into an incumbent's product line.

European companies that relocated their HQ abroad

Company	Founded in	Moved to	Year
To investigate in step 1			

Working hypothesis for the thesis

Emerging from the annex, **to be tested against the scored data, not assumed**: Switzerland generates world-class Physical AI technology through ETH and EPFL spin-offs, but value capture migrates abroad — through acquisition by foreign strategics — before those companies reach independent scale. If the scored data supports it, the interesting question becomes which of the 26 has the field traction to break that pattern.